OPERATOR HANDBOOK

N0JCG Gateway

Installation, configuration, operation, and recovery

RELEASE	0.1.9	**PUBLICATION**	August 2026
PLATFORM	Ubuntu 24.04 LTS	**AUDIENCE**	Operators and maintainers

Open radio systems, engineered as one platform.

Contents

1. About N0JCG Gateway

N0JCG Gateway is a Linux radio-station appliance and browser dashboard. It combines live APRS reception, a receive-only APRS-IS iGate, Winlink RMS Packet, a LAN Winlink Post Office, Ecowitt weather data, system readiness, and links to other N0JCG operational applications.

The ROC presents observed state. A green service label means the underlying service is running and its current health evidence is good; it is not a claim about antenna coverage or guaranteed message delivery.

What is included

- N0JCG Gateway dashboard and JSON APIs.
- RTL-SDR APRS reception with Dire Wolf.
- Receive-only APRS-IS forwarding and internet-only iGate identification.
- APRS frame history and optional aprs.fi station map.
- 1200-baud Winlink RMS Packet integration using Dire Wolf and LinBPQ.
- LAN Network Post Office access for Winlink Express.
- Ecowitt GW1100/WS90 weather monitoring.
- Protected administrative controls and privacy-safe diagnostics.
- Configurable launch links and health summaries for independently hosted N0JCG Air Traffic Center and N0JCG Scanner systems.

What is not included

- A coordinated radio frequency, Winlink gateway authorization, or Amateur Radio license.
- N0JCG Air Traffic Center or N0JCG Scanner application code.
- VARA FM, digipeating, or browser-controlled RF test transmission.
- A guarantee that every radio, USB audio interface, or SDR uses the same device name or audio level.

2. Safety, licensing, and privacy

N0JCG Gateway is intended for use by a qualified Amateur Radio operator. The operator remains responsible for station licensing, callsign use, frequency coordination, permitted modes, transmit power, identification, RF exposure, and local regulations. Winlink RMS authorization does not assign a frequency.

Before enabling any transmit-capable service:

1. Coordinate the channel locally and record the result.
2. Confirm the radio and DigiRig cable pinout.
3. Use a dummy load for PTT and audio-level tests that do not require an over-the-air peer.
4. Verify that one process owns the radio audio and PTT interfaces.
5. Begin at the minimum practical transmit power.
6. Confirm that PTT releases after every test and failure condition.

Keep passwords, passcodes, API keys, exact private coordinates, and remote access credentials outside the repository. Protected settings belong in `/etc/n0jcg/` with restrictive permissions. Mutable ROC state belongs in `/var/lib/n0jcg-roc/`.

3. System requirements

Supported host

- Ubuntu 24.04 LTS, x86-64.
- A dedicated local account named `n0jcg`.
- A wired or wireless LAN connection with a stable address or DHCP reservation.
- Python 3.11 or newer.
- Internet access for APRS-IS, aprs.fi, Winlink CMS, licensing, and updates.

Optional hardware by service

Service	Hardware
APRS RX iGate	RTL-SDR with a stable EEPROM serial and a 2 m antenna
Winlink RMS Packet	Supported VHF radio, DigiRig Mobile, radio-specific cable, antenna
Weather	Ecowitt GW1100 gateway and compatible outdoor sensor such as WS90
Air Traffic Center	Independently installed N0JCG Air Traffic Center node
Scanner	Independently installed N0JCG Scanner node

Use a powered USB hub when the host cannot reliably power all attached devices. Identify RTL-SDRs by EEPROM serial rather than Linux device number.

4. Install the ROC dashboard

The release archive contains the complete repository snapshot. Install it at the supported location:

```
sudo apt update
sudo apt install -y git python3 curl
sudo install -d -o n0jcg -g n0jcg /home/n0jcg/sdrdev
cd /home/n0jcg/sdrdev
tar -xzf N0JCG-ROC-v0.1.1.tar.gz
mv N0JCG-ROC-v0.1.1 N0JCG-ROC
cd N0JCG-ROC
```

Validate the source before installing the service:

```
./tools/validate.sh
./tools/install_base_tools.sh --check-only
```

Install the declared radio-tool baseline only after reviewing the package list. Then install the dashboard service:

```
./tools/install_base_tools.sh
./tools/install_systemd_service.sh
```

Open `http://ROC_IP/` from a browser on the same network. The default service listens on TCP port 80.

Expected result

`n0jcg-roc.service` is active, the dashboard loads, and `/api/health` returns JSON. Radio services may still show waiting or unavailable until configured.

5. First-run configuration

Station overview

Open **Configuration** and set the Station overview title. Include the operator's Amateur Radio callsign and a useful public location or description, for example `W1ABC · County EOC`. Avoid a private street address.

Registration and evaluation mode

An unregistered installation operates its data services for a five-minute evaluation period. When that period expires, APRS, Weather, and Winlink services stop and dashboard data pauses. **Restart Trial** becomes available only after expiration.

Enter a valid N0JCG Gateway license serial in **Product registration** to enable continuous operation. After successful activation the trial label and restart button are hidden. Registration does not configure or authorize RF services.

Independent application connections

In **Application connections**, configure Air Traffic Center and Scanner separately. Each application has its own enable switch, host, and port, so they may run on different computers. The ROC creates direct launch links and reads only the application health data needed for its summary cards.

6. Dashboard tour

Station overview and live summary

The top of the dashboard shows the configured station title, ROC health, and summary cards for APRS, Winlink, Weather, and connected applications. A service fault should be investigated before relying on the station unattended.

APRS activity

Latest APRS frame shows the newest decoded frame, its RF or Internet origin, and its UTC timestamp. Open it to view the full sortable history in pages of 25. Each stored frame keeps its own timestamp; the list is not a live packet sniffer.

The optional APRS map displays callsigns heard by this ROC during the most recent 24 hours. It requests current positions from aprs.fi and uses the APRS symbol carried by the station report. Configure an aprs.fi API key and enable the integration under **Configuration**. Disabling the switch stops requests without deleting the saved key.

Winlink Gateway

The Winlink section reports:

- gateway callsign, mode, frequency, and CMS state;
- current radio link and connected remote callsign;
- sessions, success rate, bytes, and throughput during the last 24 hours;
- recent session results and duration; and
- BPQMail Pending, Delivered, Received, and Archived message counts.

Pending means mail still queued for delivery. **Delivered** means the local store records successful forwarding or pickup. **Received** identifies inbound mail records. **Archived** is a retained state, so a received message may also be counted as archived; the four values are operational categories rather than a single partition of all records.

Weather monitor

Weather cards show the GW1100 connection plus temperature, humidity, pressure, wind, rain, solar, UV, and sensor status when available. The observed timestamp and stale label distinguish a working collector from an old reading.

System readiness

System readiness shows host resources, attached hardware, APRS frame totals, aircraft, P25 voice calls, and VHF/UHF locks where connected applications provide those metrics. A remote-application metric is unavailable when its connection is disabled or its API cannot be reached.

7. Configure APRS RX iGate

APRS reception uses an RTL-SDR and Dire Wolf on 144.390 MHz in North America. The DigiRig audio interface is reserved for Winlink.

1. Set a unique RTL EEPROM serial and record it.
2. Copy `config/aprs-igate.env.example` to `/etc/n0jcg/aprs-igate.env`.
3. Enter the iGate callsign and APRS-IS passcode.
4. Protect the file with owner-only permissions.
5. Install and start `n0jcg-aprs-rx.service`.

Example protection:

```
sudo install -d -m 0755 /etc/n0jcg
sudo install -m 0600 config/aprs-igate.env.example /etc/n0jcg/aprs-igate.env
sudo editor /etc/n0jcg/aprs-igate.env
sudo systemctl daemon-reload
sudo systemctl enable --now n0jcg-aprs-rx.service
```

The iGate path is receive-only. Internet identification packets do not key a radio. Confirm APRS-IS login verification in the service journal, then confirm a locally decoded RF frame before treating the iGate as commissioned.

Passive digipeater survey

The ROC includes a receive-only digipeater survey at the bottom of the APRS activity panel. It reviews the recent RF frame history (72 hours by default) and reports packets whose path contains a digipeater hop marked with `*`. Common path aliases such as `WIDE`, `TRACE`, `RELAY`, `NCA`, and `SS` are ignored, and APRS-IS-only frames are never counted as RF observations.

This is a passive planning tool: it does not transmit, query APRS-IS, or prove that a digipeater is currently reachable. A result shows that the ROC heard a frame after a digipeater had used it; an empty result means no qualifying hop was decoded during the survey window. The displayed distance is the distance from the decoded source station to the ROC, not the digipeater's location.

Use the survey before enabling any digipeater function. Coordinate a local frequency and path policy first, and treat the survey as supporting evidence for coverage planning rather than authorization to transmit.

8. Configure Ecowitt weather

Give the GW1100 a stable LAN address. Copy `config/weather.env.example` to `/etc/n0jcg/weather.env`, replace `WEATHER_GATEWAY_IP`, and install or restart `n0jcg-weather.service`.

```
sudo install -m 0600 config/weather.env.example /etc/n0jcg/weather.env
sudo editor /etc/n0jcg/weather.env
sudo systemctl enable --now n0jcg-weather.service
journalctl -u n0jcg-weather.service -n 30 --no-pager
```

The dashboard should show a current observation time and identify the outdoor sensor. Pressure may be reported as absolute or relative depending on the gateway data. Calibrate relative pressure in the Ecowitt system, not by silently changing the ROC display.

CWOP upload

The GW1100 does not upload directly to CWOP. The ROC converts the normalized GW1100 observation into an APRS weather report and sends it to CWOP through APRS-IS; this path never keys a radio. After CWOP assigns the station ID, protect the following values in `/etc/n0jcg/weather.env`:

```
CWOP_STATION_ID=CWxxxx
# CWOP weather stations use the APRS-IS unverified passcode.
CWOP_APRS_PASSCODE=-1
CWOP_LATITUDE=38.800833
CWOP_LONGITUDE=-105.200167
CWOP_INTERVAL_SECONDS=300
```

Alternatively, unlock Protected operator controls in the ROC Configuration panel, enter the CWOP station ID, station coordinates, and upload interval, and save. The panel writes `/var/lib/n0jcg-roc/cwop.json`; this is the preferred path for other operators because it does not require editing a service environment file. The upload interval is constrained to 5 minutes through 1 hour. Then enable the uploader:

```
sudo systemctl enable --now n0jcg-cwop-weather.service
journalctl -u n0jcg-cwop-weather.service -n 30 --no-pager
```

Confirm the station in the CWOP search tool after 5–15 minutes. The service is safe to leave installed while unconfigured: it remains active but skips uploads until the operator enables a complete CWOP configuration.

9. Commission Winlink RMS Packet

Do not begin this section until the gateway has Winlink authorization and a locally coordinated frequency.

Copy `config/winlink-rms.env.example` to `/etc/n0jcg/winlink-rms.env`. Enter the assigned identities, CMS password, unique management password, grid square, coordinated frequency, service hours, transmitter power, antenna height above local ground, antenna gain, and antenna direction. Keep `WINLINK_PTT_ENABLED=0` during initial setup.

Generate the runtime files:

```
sudo deploy/install_linbpq.sh --install
sudo deploy/configure_winlink_rms.sh
```

Use the preflight and dummy-load procedures in `docs/WINLINK_RMS_COMMISSIONING.md`. Confirm receive audio, PTT polarity, transmit audio, clean release, local AX.25 connection, CMS connectivity, and the Winlink channel report before unattended operation.

The service names are:

- `n0jcg-winlink-modem.service` for the Dire Wolf modem path;
- `n0jcg-winlink-rms.service` for LinBPQ and the RMS application.

LAN Winlink Post Office

The optional Network Post Office uses TCP port 8772 and does not key the radio. Enable it in the protected Winlink environment, regenerate the configuration, and restart LinBPQ. In Winlink Express choose **Network Post Office**, enter the ROC host and port 8772, and enable sending from the Outbox. Complete the user's operator name and CMS credentials in LinBPQ Mail Management before relying on automated CMS polling.

10. Protected operator controls

Configure the administrator password from a local terminal or SSH session:

```
cd /home/n0jcg/sdrdev/N0JCG-ROC
sudo deploy/configure_operator_controls.sh
```

The password must contain at least 10 characters. The script stores a salted PBKDF2-SHA256 record and a random session-signing secret, not the plaintext password. Existing browser sessions are invalidated whenever the password is changed.

After login, the dashboard can:

- restart the Winlink RMS and modem services;

- enter or leave Winlink maintenance mode;
- test the network-only CMS connection when no RF link is active; and
- download privacy-safe diagnostics.

Administrator sessions expire after 30 minutes. The controls intentionally do not provide a command shell, arbitrary service names, or an RF test button.

11. Routine operation

At the start of an operating period:

1. Confirm the dashboard health label is operational.
2. Check that expected services are active and application links are reachable.
3. Review APRS and Winlink last-activity timestamps.
4. Confirm Weather observations are current.
5. Investigate failed Winlink sessions and abnormal service restarts.

Before planned maintenance, use Winlink maintenance mode or stop the affected service. Avoid testing CMS connectivity during an active RF transfer.

12. Troubleshooting

Dashboard unavailable

```
systemctl status n0jcg-roc.service --no-pager
journalctl -u n0jcg-roc.service -n 50 --no-pager
curl -fsS http://127.0.0.1/api/health
```

Confirm the host address and local firewall. The default dashboard port is 80.

APRS service active but no frames

- Confirm the expected RTL serial is present with `rtl_test`.
- Confirm no other process owns the SDR.
- Review `journalctl -u n0jcg-aprs-rx.service`.
- Inspect the bounded diagnostic WAV produced by the listener when enabled.
- Verify frequency, mode, gain, antenna, and feedline before changing decoder parameters.

Winlink connection fails

- Review both Winlink service journals.
- Confirm radio volume, DigiRig capture level, PTT, narrow/wide mode, and coordinated frequency.
- Check current RF-link state before running a CMS test.
- Use a smaller message to separate link setup from long-transfer reliability.

Weather data stale

- Ping the GW1100 from the ROC.
- Confirm its address in `/etc/n0jcg/weather.env`.

- Check that the gateway and ROC clocks are synchronized.
- Review the collector journal for HTTP timeouts or missing fields.

13. Backup, update, and recovery

Back up these local locations before an upgrade:

- `/etc/n0jcg/` for protected service configuration;
- `/var/lib/n0jcg-roc/` for ROC settings, registration, and audit data; and
- `/var/lib/n0jcg-winlink/` for LinBPQ and BPQMail state.

Do not overwrite these directories with files from a release archive. Validate a new release before restarting services. Preserve the previous archive and a copy of the service configuration so the installation can be rolled back.

To collect a support snapshot, use the authenticated diagnostics control or run the read-only status and journal commands in this guide. Remove callsigns, addresses, message metadata, and network details before sharing a diagnostic outside the station team.

14. Release verification and support

Compare the downloaded archive with `SHA256SUMS.txt` before installation. The release is published from the `main` branch at:

```
https://github.com/bakstaaj/N0JCG-ROC
```

When reporting an issue, include the release version, Ubuntu version, affected service, relevant redacted journal lines, hardware model, and the exact expected versus observed behavior. Never include passwords, passcodes, license serials, private keys, or message contents.

N0JCG Open Radio Platform — Open radio systems, engineered as one platform.